

SNEHADRI DUTTA

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PROFESSIONAL SUMMARY

Senior AI Engineer with 5+ years of experience specializing in the design, deployment, and operation of agentic AI systems and scalable RAG architectures. Expert in building end-to-end Generative AI solutions—from initial prototyping to production-grade workloads on AWS. Highly proficient in agentic workflows (LangGraph/LangChain), function calling, and tool orchestration using frontier models like Llama 3. Proven track record in applying rigorous evaluation techniques and observability practices to ensure model reliability, safety, and performance. A strong advocate for software engineering excellence, combining deep Python/FastAPI expertise with a background in distributed systems and C#/.NET to build robust, secure, and cost-optimized AI platforms.

WORK EXPERIENCE

Senior ML Engineer

Nerve Solutions, Mumbai | October 2025 - Present

Tech Stack: Llama 3.x, Ollama, LangGraph, LangChain, Hugging Face, Agent Flows (ReAct, Planner-Executor), RAG Architectures, Multi-step Reasoning, Prompt Engineering (CoT, Graph-of-Thought), MCP Integration, Tool Calling, Vector DBs (FAISS, ChromaDB), Embeddings, FastAPI, Cloud AI Services, Python.

- Architected a production-grade Financial Intelligence Agent using a Router-Synthesizer pattern. Designed autonomous agents capable of multi-step reasoning, utilizing lightweight models for intent classification and frontier models for complex data synthesis.
- Own the full lifecycle of AI workloads, deploying scalable agentic systems into AWS-based environments using FastAPI. Established robust CI/CD pipelines, automated testing, and versioning to ensure seamless production releases.
- Engineered sophisticated RAG pipelines with vector search (FAISS/ChromaDB) and optimized embeddings. Developed dynamic function-calling layers that integrate LLMs with real-time REST APIs and enterprise data platforms.
- Implemented comprehensive observability practices, including logs, metrics, and traces, to monitor model behavior and system health. Utilized Groq LPU to optimize latency and performance, ensuring sub-second response times for live market queries.
- Applied advanced LLM evaluation techniques to assess quality and safety. Refined system reliability through extensive Prompt Engineering (CoT, ReAct) and fine-tuning, ensuring compliance with rigorous financial industry standards.

Developer Analyst

Nerve Solutions, Mumbai | March 2021 – September 2025

Tech Stack: C#/.Net, NLP, RESTful API, Multithreading, LINQ, Microservices, System Architecture, SQL/MongoDB.

- Designed and optimized high-performance backend algorithms and data pipelines using C# (.NET), ensuring system stability, scalability, and throughput in a low-latency production environment.
- Built, deployed, and supported a machine learning-based recommendation engine, analysing complex user behavioral datasets to deliver actionable portfolio insights.
- Developed real-time risk monitoring and alerting systems to support rapid incident detection, empowering proactive troubleshooting and long-term platform reliability.
- Led data processing, API development, and ML model integration workflows, while mentoring a development team to deliver production releases that contributed to a 30% increase in client growth.

EDUCATION

Bachelor of Technology in Aeronautical

Hindustan Institute of Technology and Sciences, Chennai

Graduated: 2019

SKILLS

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|-------------|-----------------|-------------------------|--------------|
| • LangGraph | • Generative AI | • Large Language Models | • Embeddings |
| • LangChain | • Python | • FastAPI | • GCP |

RELEVANT PROJECTS

- Developed CodeLens, an AI-powered codebase Q&A assistant utilizing a Router-Synthesizer architecture and LangGraph to provide precise technical answers grounded in file and line citations. Integrated FastAPI, Qdrant, and Tree-sitter to implement AST-based code chunking and hybrid search, while ensuring robustness through production-grade controls like rate limiting.
- Implemented an algorithm using OpenCV to capture and analyze individual faces through a pre-trained VGG-16 model, making it ideal for integration into surveillance systems for effective person tracking.
- Engineered a Convolutional Neural Network model with OpenCV to assess emotions on detected faces within a frame, presenting corresponding emojis. Applicable in theatres or shows, this model can track audience emotions, enhancing engagement and feedback.

LANGUAGES

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|-----------|---------|-----------|----------------|
| • English | • Hindi | • Bengali | • Deutsche(A1) |
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